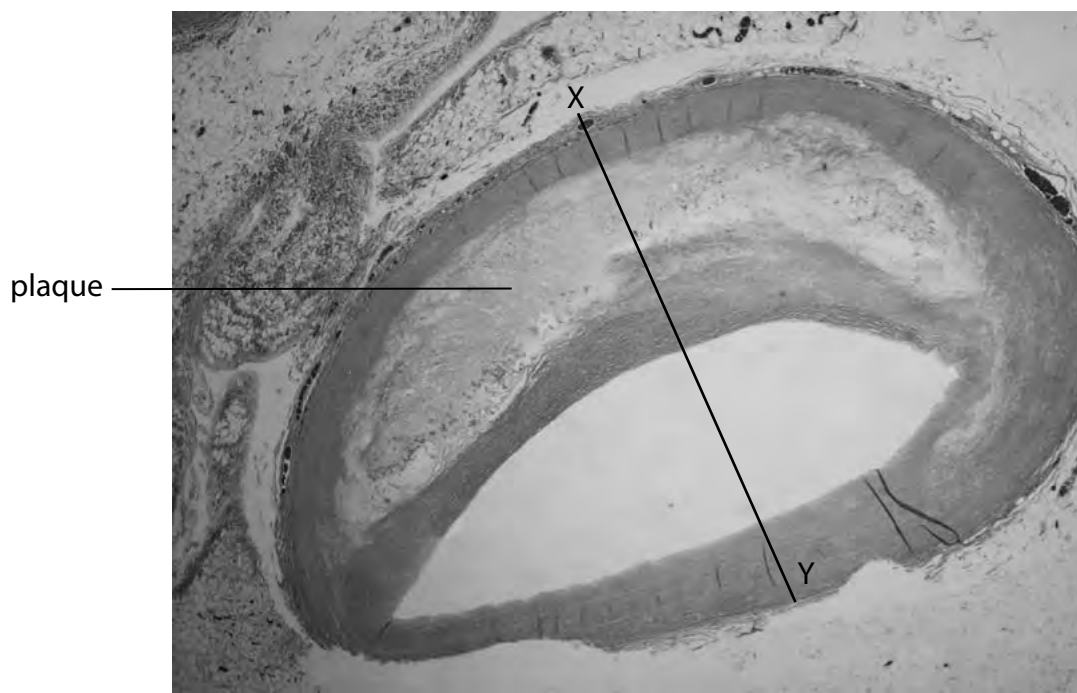


8 Many studies have linked the development of atherosclerosis with cardiovascular disease (CVD).

- (a) The photograph shows a section through an artery with a plaque (atheroma) from a patient with atherosclerosis.



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Calculate the percentage increase in the thickness of the artery wall where the plaque is located.

Take your measurements along the line labelled X–Y.

(3)

Answer%

(b) Explain how atherosclerosis can result in damage to the heart muscle.

(4)

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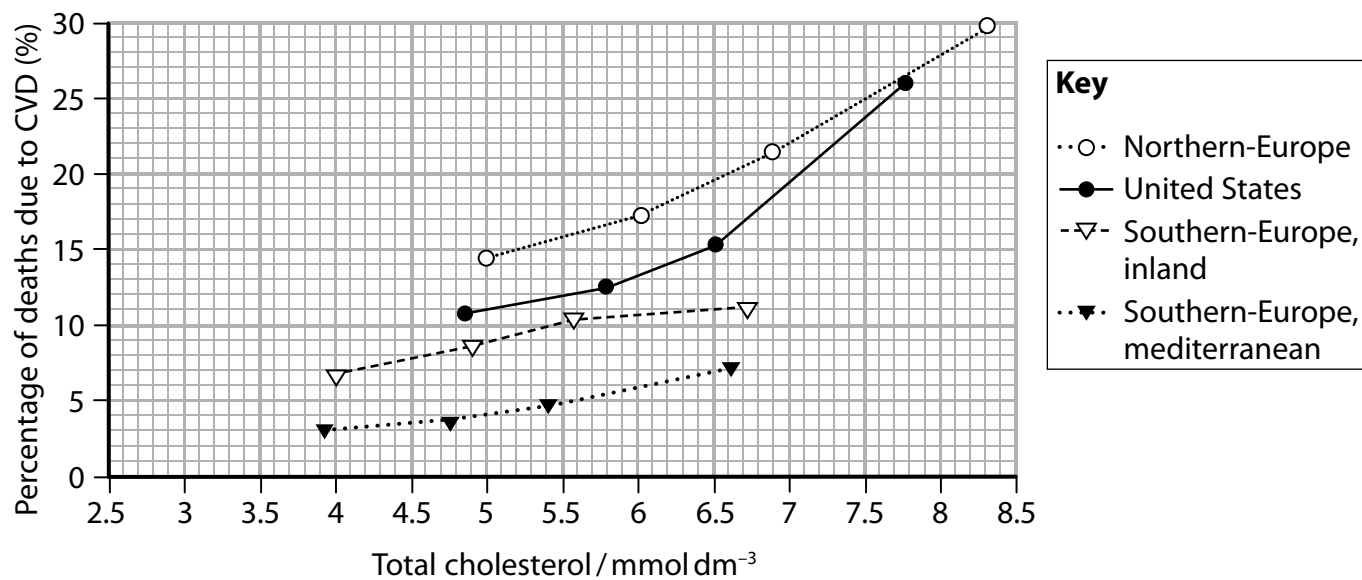
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*(c) Cholesterol is transported in the blood as lipoproteins LDL and HDL.

In one study, the relationship between total blood cholesterol and the risk of death from CVD was investigated.

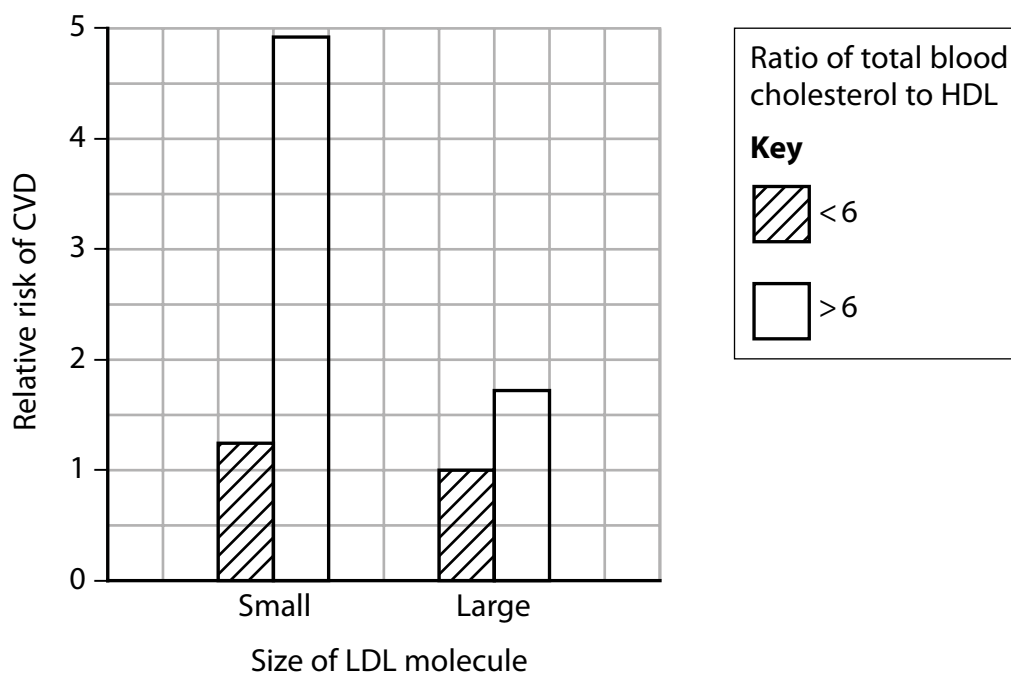
The results are shown in the graph.



Graph 1

In another study, the effect of the size of LDL and the ratio of total blood cholesterol to HDL on the relative risk of CVD was investigated.

The results are shown in the graph.



Graph 2

Assess the contribution of lipoproteins to the risk of developing CVD. Use the information in the graph to support your answer.

(6)

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(Total for Question 8 = 13 marks)

TOTAL FOR PAPER = 80 MARKS